Database QAP 1

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Benefits of using a database for the bookstore:

- Reduction of Data Redundancy – With a database, information is stored in normalized tables so that author names, books, customer details etc are stored only once.
- Improved Efficiency and Accuracy – Allows staff to quickly retrieve general and specific information, such as a customers order or specific information about an author.
- Scalability and Growth – As the bookstore's inventory grows, the database can easily and efficiently scale with it. The uniform model allows ease of storing information.

- As customers order books, the bookstore can easily track which books sell the most and adjust their inventory accordingly.

Authors

AuthorID (PK)	Name	BirthPlace
001	George Orwell	India
002	Steving King	United States
003	J.R.R Tolkien	South Africa

- 1NF: All attributes store a single value with no repeating groups
- 2NF: The primary key is AuthorID so no partial dependencies exist
- 3NF: All non-key attributes depend only on AuthorID

Books

Authors → Books: 1:M

BookID(PK)	Title	ISBN	Year	Price	AuthorID(FK)
A1	The Shining	9780307743657	1977	16.99	002
A2	1984	9780155658110	1949	14.84	001
A3	Lord of the Rings	9780007123827	1954	18.50	003

The foreign key AuthorID is unique to each BookID. Titles, years, and prices can be repeated. One author can write many books, but each book only has 1 Author which is a 1:M relationship.

- 1NF: All attributes store a single value with no repeating groups
- 2NF: The primary key is BookID so no partial dependencies exist
- 3NF: All non-key attributes depend only on BookID

Genres

GenreID(PK)	Name
B1	Fantasy
B2	Dystopian
B3	Horror
B4	Fiction
B5	Science-Fiction

- 1NF: Each genre has a single name with no repeating lists
- 2NF: Primary key is GenreID
- 3NF: Name depends onlyon GenreID

Book Genres

Books → Genres: M:M

BookID(FK)	GenrID(FK)
A1	B3
A2	B2, B5
A3	B1, B4

Book Genres Table is a junction table that creates a composite key of BookID and GenreID. Each book can have many genres, and each genre can have many books, which makes this a M:M relationship.

- 1NF: All attributes store a single value with no repeating groups
- 2NF: Composite PK avoids partial dependencies (BookID and GenreID)
- 3NF: There are no other attributes so this complies with 3NF requirements

Customers

CustomerID(PK)	**Name**	**Email**	**Address**
AA-1	Boby Brown	Boby@email.com	123 Maple St
AA-2	Jimmy Jones	jimcool@email.com	456 Water Rd
AA-3	Susan Simpson	flowersimp@email.com	890 Rock Ln

- 1NF: Each attribute has one value per field
- 2NF: Primary key CustomerID is used so no partial dependencies exist
- 3NF: The Email and Address depend only on CustomerID, Names can be repeated but are also attached to the CustomerID ensuring no duplicates

Orders

Customer → Order: 1:M

OrderID(PK)	CustomerID(FK)	OrderDate	Status
1-BB	AA-2	Sept 16	Shipped
2-BB	AA-3	Oct 1	Pending
3-BB	AA-1	Sept 5	Out for Delivery

A customer can place many orders, but each order belongs to one customer. This is a 1:M relationship

- 1NF: Each attribute has one value per field 2NF: Primary key OrdID is
- used so no partial dependencies exist 3NF: CustomerID is a foreign key
- but non-key attributes depend only on OrderID

Order Details

Books → Orders: M:M

OrderID(FK)	BookID(FK)	Quantity
2-BB	**A1**	**1**
2-BB	A3	1
1-BB	A1	2
3-BB	A2	1

Order Details table is a junction table created with a composite key of OrderID and BookID with a quantity value. Books can be in multiple orders and vice versa, a M:M relationship

- 1NF: Each attribute has one value per field
- 2NF: Primary key OrdID is used so no partial dependencies exist
- 3NF: CustomerID is a foreign key but non-key attributes depend only on OrderID

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